Frobenius Minimality, Minimum-Layer Uniqueness, and an Infinite Family of Huneke–Wiegand Counterexamples

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Provenance. Son Pham discovered the counterexample; Professor Craig Huneke independently verified it. This paper contributes certified Frobenius minimality, minimum-layer uniqueness, and a separate explicit infinite family.

Abstract

Son Pham publicly identified a counterexample to the Huneke–Wiegand conjecture in the class of two-generated monomial ideals over symmetric numerical semigroup rings, subsequently verified by Professor Craig Huneke. We preserve that discovery priority and answer two different questions: how small can such an example be, and does the same semigroup/ideal class contain an explicit infinite family? We prove by proof-carrying exact computation that the least Frobenius number is 181, attained by Pham's semigroup at ideal shift 14. First, an implementation of the complete Blanco–Rosales tree exhausts all 48,954 symmetric semigroups with Frobenius number below 69 and all 1,503,391 associated gap cases. An independent Boolean route checks 1,156 fixed-pair UNSAT certificates for the same range. We then introduce a one-hot shift-selector CNF that asks one existence question per Frobenius number. Its DRAT certificates exclude every odd value from 69 through 179, while the formula at 181 is satisfiable and decodes exactly to the public semigroup and shift. A separately generated fixed-pair formula returns the same model. An independent audit rehashes all 228 files in the minimality search, totaling 760,081,739 bytes, rechecks both SAT models, and reproduces the strict-order manifest with no mismatch. Complete theorem-tree enumeration at $F = 69, 71, 73, 75$ provides an object-level cross-check on another 79,790 semigroups and 2,933,163 gap cases. A second projected enumeration proves that shift 14 is the only feasible shift at $F = 181$ and that the public membership vector is the only semigroup there. Both terminal DRAT proofs are freshly rechecked in an independent 12-file audit. Thus Pham's pair is the unique normalized pair attaining the Frobenius minimum in this class. Finally, for every integer $p \geq 4$ we construct a symmetric numerical semigroup Γ_p of multiplicity $24p$, Frobenius number $78p - 1$, conductor $78p$, and embedding dimension $11p$, for which $I_p = (t^{24p}, t^{30p})$ is nonprincipal and rigid. This family theorem is deductive: seven exact interval-sum identities prove closure, symmetry, and $D_s = E_s + E_s$; a finite sweep through $p = 300$ and an independently implemented audit are supporting checks. The results remain confined to two-generated monomial ideals over symmetric numerical semigroup rings and do not classify arbitrary Gorenstein domains and modules.

1 Introduction

Let R be a one-dimensional Gorenstein local domain and let M be a nonzero finitely generated non-free module. The Huneke–Wiegand conjecture predicted nontrivial torsion in $M \otimes_R \mathrm{Hom}_R(M, R)$ [1]. For a numerical semigroup $\Gamma \subseteq \mathbb{N}$, the semigroup ring $\Bbbk[t^\Gamma]$ is Gorenstein exactly when Γ is symmetric. This specialization converts part of the module problem into finite additive combinatorics. García-Sánchez and Leamer proved broad positive classes and reported a NumericalSgps check for every symmetric semigroup with Frobenius number below 69 [2].

In 2026, Son Pham published a symmetric numerical semigroup and a two-generated monomial ideal for which the predicted torsion is absent [5]. Professor Craig Huneke supplied an independent verification archived with the public candidate record. That example has Frobenius number 181. The discovery settles the original conjecture negatively, but does not by itself say whether 181 is small, minimal, or typical.

This paper reports computer-assisted classification theorems and a deductive family theorem. Claims are labeled **[MV]** when they are verified by exact machine computation and independently checked artifacts, and **[D]** when derived in the text. No conjectural claim is needed for the main theorem.

Contributions.

1. We give a self-contained finite-window reduction and a direct proof that the selector CNF is equisatisfiable with existence of a rigid two-generated monomial ideal.
2. We reproduce the published $F < 69$ boundary by two independent complete routes: the Blanco–Rosales tree and checked DRAT proofs.
3. We certify UNSAT for every odd $F = 69, 71, \ldots, 179$, then recover the exact public model at $(F, s) = (181, 14)$.
4. We independently rehash and inspect the complete search artifact set, and cross-check the first four new odd Frobenius values by explicit theorem-tree enumeration.
5. We classify the complete minimum layer by projected enumeration: the only feasible shift at $F = 181$ is 14, and the public semigroup is the only membership vector at that shift.
6. We identify two structural boundaries: the public semigroup lies outside the generalized arithmetic-sequence positive family, and the visible fixed-offset generator blocks do not extend to a naive parametric family.
7. We use the failed fixed-width ansatz to derive and prove an explicit infinite family for every integer $p \geq 4$, with a separate exact implementation and independent audit.

2 The finite semigroup problem

Definition 2.1. A *numerical semigroup* is a cofinite additive submonoid $\Gamma \subseteq \mathbb{N}$. Its Frobenius number is $F(\Gamma) = \max(\mathbb{N} \setminus \Gamma)$. It is *symmetric* when

$$n \in \Gamma \iff F(\Gamma) - n \notin \Gamma \qquad (0 \leq n \leq F(\Gamma)).$$

Fix a symmetric semigroup Γ with Frobenius number F and a positive gap $s \notin \Gamma$. Normalize the corresponding two-generated relative ideal to $(0, s)$. Define

$$E_s = \{n \in \mathbb{N} : n, n+s \in \Gamma\},$$
$$D_s = \{n \in \mathbb{N} : n, n+s, n+2s \in \Gamma\}.$$

The García-Sánchez–Leamer reduction identifies the counterexample condition with

$$D_s = E_s + E_s. \tag{1}$$

Equivalently, every length-two arithmetic sequence of step s in Γ decomposes as a sum of two length-one sequences [2, 4].

Lemma 2.2 (Automatic inclusion, [D]). *For every numerical semigroup Γ and every positive integer s, $E_s + E_s \subseteq D_s$.*

Proof. If $a, b \in E_s$, then $a, a+s, b, b+s \in \Gamma$. Closure gives $a+b \in \Gamma$, $a+b+s = a+(b+s) \in \Gamma$, and $a+b+2s = (a+s)+(b+s) \in \Gamma$. Hence $a+b \in D_s$. □

Lemma 2.3 (Finite window, [D]). *Equation (1) holds if and only if $D_s \cap [0, 2F+1] \subseteq E_s + E_s$.*

Proof. Only the reverse inclusion needs checking by Lemma 2.2. Every integer at least $F+1$ belongs to Γ, hence to E_s and D_s. Let $m = \min E_s$; then $m \leq F+1$. For every $n \geq m+F+1$, both m and $n-m$ lie in E_s, so $n \in E_s + E_s$. Since $m+F+1 \leq 2F+2$, every $n \geq 2F+2$ is automatic. The displayed finite window is therefore complete. □

Lemma 2.4 (Parity, [D]). *The Frobenius number of a symmetric numerical semigroup is odd.*

Proof. If F were even, symmetry at $n = F/2$ would require $F/2$ to belong to Γ exactly when it does not belong to Γ. □

3 The public model and independent calibration

Pham's public semigroup is

$$\Gamma = \langle 56, 57, 58, 63, 64, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83,$$
$$87, 89, 90, 93, 95, 96, 97\rangle,$$

with the ideal generated by t^{56} and t^{70}, hence normalized shift $s = 14$ [5]. An independent Singular/4ti2 calculation and a separate standard-library finite route verify $F = 181$, conductor 182, genus 91, symmetry, and equality (1) **[MV]**. The Singular route uses a 322-generator toric standard basis; both reduced ideal differences vanish. The hypersurface control $\langle 4, 5\rangle$ rejects equality with explicit residues **[MV]**.

The endomorphism overring of the ideal has value semigroup

$$v(\mathrm{End}_R(I)) = \Gamma \cup \{101, 107, 181\},$$

with Frobenius number 125, genus 88, and type 24 **[MV]**. This explains why positive criteria that require a Gorenstein endomorphism ring do not apply, but it does not imply minimality.

4 Complete reproduction below 69

For fixed odd F, irreducible numerical semigroups are exactly the symmetric ones. Blanco and Rosales construct all of them as a rooted tree [3]. Starting from

$$C(F) = \{0, (F+1)/2, (F+1)/2+1, \ldots\} \setminus \{F\},$$

a child replaces a suitable minimal generator $x > F/2$ by $F - x$. Their theorem supplies exact child conditions, completeness, and a unique parent.

Our direct implementation validates every node for closure, symmetry, genus, minimal generators, and its theorem parent. It then checks every gap by Lemma 2.3 and retains a first element of $D_s \setminus (E_s + E_s)$ whenever equality fails. The full range $F = 1, 3, \ldots, 67$ contains 48,954 semigroups and 1,503,391 gap cases; none is rigid **[MV]**. At $F = 11$ the implementation returns the six nodes listed by Blanco–Rosales, while an invalid mutation is rejected by closure at $(1, 1)$.

An independent route generates one fixed-pair DIMACS formula for each (F, s), including shifts that are forced not to be gaps. CaDiCaL emits a DRAT proof for every UNSAT result, and DRAT-trim checks each proof against the exact CNF [6, 7]. All 1,156 formulas are UNSAT and independently verified **[MV]**. A committed auditor rehashes 4,624 external files totaling 211,671,241 bytes with zero mismatch.

5 One selector formula per Frobenius value

The fixed-pair route duplicates the same semigroup constraints once per shift. We instead introduce Boolean membership variables h_n for $0 \le n \le F$ and one-hot variables q_s for $1 \le s \le F$.

5.1 Semigroup and selector clauses

The CNF contains units h_0 and $\neg h_F$, the symmetry clauses

$$(h_n \vee h_{F-n}) \wedge (\neg h_n \vee \neg h_{F-n}),$$

and closure clauses $\neg h_a \vee \neg h_b \vee h_{a+b}$ for $a + b \le F$. The selectors satisfy one at-least-one clause, all pairwise at-most-one clauses, and $q_s \Rightarrow \neg h_s$. Thus every model chooses exactly one gap.

For $0 \le n \le 2F + 1$, guarded Tseitin clauses impose, under the selected q_s,

$$\begin{aligned} e_n &\longleftrightarrow h_n \wedge h_{n+s}, \\ d_n &\longleftrightarrow h_n \wedge h_{n+s} \wedge h_{n+2s}, \end{aligned}$$

where h_j is the constant true for $j > F$. For each unordered decomposition $n = a + b$, a variable $p_{n,a,b}$ is equivalent to $e_a \wedge e_b$. Finally,

$$d_n \Rightarrow \bigvee_{a+b=n} p_{n,a,b}.$$

The automatic inclusion from Lemma 2.2 need not be duplicated in the formula.

Proposition 5.1 (Selector equisatisfiability, [D]). *For positive odd F, the selector CNF is satisfiable if and only if there exist a symmetric numerical semigroup Γ with Frobenius number F and a gap s satisfying $D_s = E_s + E_s$.*

Proof. The semigroup clauses decode exactly a symmetric semigroup of Frobenius number F, and the one-hot clauses decode a unique gap. Because exactly one guard is true, all e_n and d_n equal the definitions for that gap. The p equivalences and final disjunctions impose $D_s \subseteq E_s + E_s$ through $2F+1$. Lemmas 2.2 and 2.3 give global equality. Conversely, from a rigid pair assign h and q from the pair, assign e, d by their definitions, and assign each p by its conjunction. Every clause is then true. □

The selector encoding passes two adversarial calibrations [**MV**]. Every odd $F \le 67$ is UNSAT with an accepted proof, while $F = 181$ is SAT. The latter formula has 34,397 variables and 363,912 clauses, selects $s = 14$, and decodes exactly to the public membership vector. A newly generated fixed-pair formula at $(181, 14)$ returns the same vector.

6 The certified search and minimality theorem

We scan odd F in strict increasing order. Table 1 summarizes the closed run.

Table 1: Selector-CNF minimality search.

Quantity	Value
Odd Frobenius values requested	57 (69 through 181)
UNSAT with accepted DRAT proof	56 (69 through 179)
Validated SAT models	1 (181, shift 14)
UNKNOWN or timeout	0
Wall time	4,303.86 seconds
CNF bytes	169,745,433
DRAT proof bytes	586,785,257
Slowest solve	$F = 175$, 218.83 seconds

Theorem 6.1 (Frobenius minimality, [MV+D])**.** *Among symmetric numerical semigroups admitting a nonprincipal two-generated monomial ideal that is a counterexample to the Huneke–Wiegand prediction, the least Frobenius number is* 181. *It is attained by Pham's public example at shift* 14.

Proof. By Lemma 2.4, only odd Frobenius values occur. Complete tree enumeration and the independent fixed-pair proof suite exclude every odd $F \le 67$. Proposition 5.1 and the accepted selector DRAT proofs exclude every odd F from 69 through 179. The validated selector and fixed-pair models at $(181, 14)$ provide existence. These cases exhaust all smaller positive Frobenius values. □

The search audit independently rehashes all 228 expected CNF, proof, solver-log, and checker-log files, totaling 760,081,739 bytes. It also reconstructs both SAT membership masks, validates closure, symmetry, and rigidity, checks strict odd ordering, and recomputes aggregate SHA-256

`0f580de2707a00fdd52e1b3c04e7767b97ce7b0a826593b119e9a49ae04da743`.

No file is missing, no hash differs, and no status marker or semantic check fails [**MV**].

7 Complete classification at the minimum

The selector search proves existence at $F = 181$ but, by itself, does not classify every model at that value. We close this gap in two projected layers. First, after discovering a valid selector model at shift s, we add the unit clause $\neg q_s$. Second, for each feasible shift we enumerate the fixed-shift formula and block each complete membership projection.

Lemma 7.1 (Projected blocker, [D]). *Let $x_1, \dots, x_k$ be distinct Boolean variables with complete assignment a. The clause*

$$B(a) = \bigvee_{a_i=1} \neg x_i \;\vee \bigvee_{a_i=0} x_i$$

is false exactly on assignment a. Its truth is independent of every variable outside the projection.

Proof. Every displayed literal is false under a. Any other complete assignment differs at some coordinate j, where the corresponding displayed literal is true. Variables absent from the clause cannot affect either statement. □

For semigroup enumeration, $B(a)$ contains one signed literal for every $h_0, \dots, h_{181}$. Consequently, it excludes one mathematical membership vector and every auxiliary extension of that vector, while preserving every different semigroup. Blocking only the positive coordinates would not have this property.

Theorem 7.2 (Minimum-layer uniqueness, [MV+D]). *Up to translation and interchange of the two monomial generators, Pham's pair is the unique nonprincipal two-generated monomial-ideal counterexample over a symmetric numerical semigroup ring with Frobenius number* 181*.*

Proof. The unblocked selector formula returns the exact public pair at shift 14. After adding $\neg q_{14}$, the formula is UNSAT and its DRAT proof is accepted, so no other shift is feasible. The fixed $(181, 14)$ formula returns the exact public membership vector. After adding its complete 182-literal blocker from Lemma 7.1, that formula is UNSAT and its DRAT proof is accepted. Thus no second semigroup exists at the only feasible shift. Translation and interchange reduce any two monomial generators to one positive normalized shift, which proves the statement. □

Table 2: Certificate summary for the complete $F = 181$ layer.

Quantity	Value
Feasible shifts	1, namely $\{14\}$
Normalized rigid pairs	1
Distinct membership vectors	1
Support terminal proof bytes	45,867,741
Fixed-class terminal proof bytes	1,608,691
Audited external files	12
Audited external bytes	63,609,504
Fresh proof-recheck time	232.88 seconds

The independent auditor reconstructs all four CNFs byte-for-byte, decodes both SAT logs, reruns closure, symmetry, minimal-generator and full-window/tail rigidity checks, and freshly checks both terminal proofs [**MV**]. It accounts for every external file and reproduces manifest SHA-256

`02d1265b75e886bc6ec693e60b367b661476170c755e38a220342396564ef5d3`.

The compact classification aggregate is

`b37d9410fa07c00f22ccc0f83796644db3b44f0dd654c06933aa5d22cb1ed788`.

8 Independent object-level checks beyond the old frontier

The theorem-tree route was continued beyond 67 without consulting SAT models. Table 3 lists the first four new odd values.

Table 3: Complete Blanco–Rosales tree cross-checks beyond the published boundary.

F	Symmetric semigroups	Gap cases	Rigid cases	Seconds
69	11,276	394,660	0	137.00
71	19,812	713,232	0	235.37
73	25,405	939,985	0	333.21
75	23,297	885,286	0	330.10
Total	79,790	2,933,163	0	1,035.68

This is not needed to trust a DRAT derivation, but it attacks the custom encoding at the object level and supplies explicit failed-rigidity witnesses for every tested gap. The routes agree.

9 From a failed ansatz to an infinite family

Landeros et al. prove the positive result for semigroups generated by a generalized arithmetic sequence $\langle a, ah+d, \ldots, ah+kd\rangle$ [4]. The public semigroup cannot lie in this class [**D**]. Its multiplicity forces $a = 56$. Since $57 \in \Gamma$ and no sum of two positive semigroup elements is below 112, the presentation forces $h = d = 1$. Since $63 \in \Gamma$, it would then include $59, 60, 61, 62$, but all four are gaps.

The seed also displays $56 = 4s$ and $181 = 13s - 1$ at $s = 14$, with generator blocks

$$4s + \{0, 1, 2, 7, 8\}, \qquad 5s + [0, s-1], \qquad 6s + \{3, 5, 6, 9, 11, 12, 13\}.$$

A predeclared exact sweep refutes the naive fixed-offset lift [**MV**]. Only $s = 14$ passes among even $s \le 100$. At $s = 16$, the generated semigroup has $F = 205$ rather than 207; at $s = 28$ the predicted Frobenius value returns but symmetry fails. This negative result prevents the minimality theorem from being overinterpreted as evidence of an immediate family.

We next retained only the structural scaffold $m = 4s$, $F = 13s-1$, the full interval $[5s, 6s-1] \subseteq \Gamma$, and rigidity at shift s. A proof-carrying Boolean search certifies that the scaffold is infeasible at $s = 16, 18$ and returns independently validated models for every even $s = 20, 22, \ldots, 40$ [**MV**]. The eleven non-seed models justified a separate extraction stage, but did not themselves prove recurrence. The first fixed-width interval formula passed three parameters and then failed: at $s = 38$, $9s + 7$ lies in D_s but not in $E_s + E_s$. For every later parameter in that formula, carried sums from the first block end at residue 6, while the next summand block begins at residue at least 8. This uniform obstruction suggests widening the first interval with the parameter.

9.1 The parametric blocks

Fix an integer $p \geq 4$ and put $s = 6p$. All intervals below are integer intervals. Define subsets of $[0, s-1]$ by

$$\begin{aligned} A_p &= [0, p] \cup [3p, 4p-2], \\ B_p &= ([p+1, 3p-1] \setminus \{2p-1\}) \cup \{4p\} \cup [5p-1, 6p-1], \\ C_p &= [0, 2p] \cup [3p, 5p-2]. \end{aligned}$$

Define $\Gamma_p \subseteq \mathbb{N}$ by the membership blocks

$$\{0\}, \quad 4s + A_p, \quad [5s, 6s-1], \quad 6s + B_p, \quad 8s + C_p, \quad [9s, 13s-2], \quad [13s, \infty), \tag{2}$$

with every unlisted nonnegative integer below $13s$ a gap.

For subsets of $[0, s-1]$, let $\mathrm{low}(X+Y)$ denote the sums below s, and let $\mathrm{carry}(X+Y)$ denote the sums at least s after subtraction of s.

Lemma 9.1 (Residue identities, [D]). *For every $p \geq 4$,*

$$\begin{aligned} &\mathrm{low}(A_p + A_p) = C_p, && \mathrm{carry}(A_p + A_p) = [0, 2p-4], \\ &\mathrm{low}(A_p + B_p) = [p+1, 6p-1], && \mathrm{carry}(A_p + B_p) = [0, 4p-3], \\ &\mathrm{low}(B_p + B_p) = [2p+2, 6p-2], && \mathrm{carry}(B_p + B_p) = [0, 6p-2], \\ &\mathrm{low}(A_p + C_p) = [0, 6p-2]. && \end{aligned}$$

Moreover,

$$[0, 2p-4] \cup [p+1, 6p-1] = [0, 6p-1], \tag{3}$$

$$[0, 4p-3] \cup [2p+2, 6p-2] = [0, 6p-2]. \tag{4}$$

Proof. Split A_p into its two displayed intervals and split the punctured component of B_p at $2p-1$. Pairwise interval addition, followed by separating sums below and above $s = 6p$, gives the seven displayed identities. The singleton $4p$ fills the endpoint removed from the middle of B_p. All interval joins are immediate except (3): its two pieces meet exactly when $2p-3 \geq p+1$, equivalently $p \geq 4$. For (4), the overlap condition is $4p-2 \geq 2p+2$, which also holds for $p \geq 4$. □

Theorem 9.2 (Infinite family, [D]). *For every integer $p \geq 4$, the set Γ_p in (2) is a symmetric numerical semigroup with multiplicity $24p$, Frobenius number $78p-1$, conductor $78p$, and embedding dimension $11p$. Let*

$$R_p = \Bbbk[t^{\Gamma_p}]_{(t^n : n \in \Gamma_p \setminus \{0\})}, \qquad I_p = (t^{24p}, t^{30p}) R_p.$$

Then R_p is Gorenstein and I_p is nonprincipal and rigid. Hence the pairs (R_p, I_p) form an infinite family of counterexamples in the two-generated monomial-ideal class.

Proof. We first prove the semigroup assertions. The only sums of positive displayed members that can stay below the conductor have starting levels among $4, 5, 6, 8$. Lemma 9.1 handles the non-full target blocks: $A_p + A_p$ produces $8s + C_p$ and part of level 9; $A_p + B_p$, $B_p + B_p$, and $A_p + C_p$ land exactly inside the declared high blocks and never produce the Frobenius residue $s-1$ in level 12. Sums involving the full level 5 block either fill a declared level or enter the conductor tail. Thus Γ_p is closed.

Conversely, the lower three blocks generate every displayed member. The identity $\mathrm{low}(A_p + A_p) = C_p$ generates level 8. The presence of $0 \in A_p$ makes $(4s + A_p) + [5s, 6s - 1]$ fill level 9, and two level-5 elements fill level 10 and residues $0, \ldots, s - 2$ of level 11; the residue $s - 1 \in B_p$ fills the remaining point. Three elements from $4s + A_p$ generate residues $0, \ldots, 3p$ of level 12, while $\mathrm{low}(B_p + B_p) = [2p + 2, s - 2]$ completes that level through $13s - 2$. Adding $4s$ to $[9s, 13s - 2]$ gives $[13s, 17s - 2]$, and $(4s + 1) + (13s - 2) = 17s - 1$. These are $4s$ consecutive generated integers, so repeated addition of the multiplicity gives the entire tail.

Therefore the multiplicity is $4s = 24p$, the conductor is $13s = 78p$, and the Frobenius number is $13s - 1 = 78p - 1$. Every member of the three lower blocks lies below $7s$, whereas a sum of two positive members is at least $8s$; all are minimal generators. Since $|A_p| = 2p$, $|B_p| = 3p$, and the full block has size $6p$, the embedding dimension is $11p$.

Symmetry is blockwise about $F = 13s - 1$. The residue-reflection complement of A_p is exactly C_p. The full level-5 block reflects to the empty level-7 block. In B_p, exactly one residue from each pair $\{r, s - 1 - r\}$ occurs: the puncture $2p - 1$ pairs with the singleton $4p$, and the remaining displayed intervals pair with their omitted intervals. The gaps below $4s$ reflect to the full range $[9s, 13s - 2]$, and 0 reflects to the gap F. Hence Γ_p is symmetric, so R_p is Gorenstein.

It remains to prove rigidity. Normalize I_p to $J_p = (1, t^s)$ and use E_s, D_s from Section 2. Direct inspection of (2) gives the nontrivial residue layers

$$
\begin{aligned}
&E_4 = A_p, && E_5 = B_p, && E_8 = C_p,\\
&E_9 = E_{10} = [0, s - 1], && E_{11} = E_{12} = [0, s - 2], &&\\
&D_8 = C_p, && D_9 = [0, s - 1], &&\\
&D_{10} = D_{11} = D_{12} = [0, s - 2]. &&&&
\end{aligned}
$$

All lower D layers are empty; in particular D_4 is empty because $A_p \cap B_p = \varnothing$. Lemma 9.1 and (3)–(4) compute the five load-bearing layers of $E_s + E_s$:

$$D_8 = \mathrm{low}(A_p + A_p), \quad D_9 = \mathrm{carry}(A_p + A_p) \cup \mathrm{low}(A_p + B_p),$$

$$D_{10} = \mathrm{carry}(A_p + B_p) \cup \mathrm{low}(B_p + B_p), \quad D_{11} = \mathrm{carry}(B_p + B_p), \quad D_{12} = \mathrm{low}(A_p + C_p).$$

At levels 13 and 14, the zero residue in A_p translates full high E blocks. At levels 15 and 16, those high blocks omit only $s - 1$, and the residues $0, 1 \in A_p$ fill both the interval and its final endpoint. Level 17 is filled by $E_4 + E_{13}$; levels 18, 19, and 20 by $E_9 + E_9$, $E_9 + E_{10}$, and $E_{10} + E_{10}$; and level 21 by $E_8 + E_{13}$ because $0 \in C_p$. Every level from 22 onward is filled by $E_9 + E_q$ with $q \geq 13$, where the conductor makes E_q full. Thus $D_s \subseteq E_s + E_s$; Lemma 2.2 gives equality.

Finally, s is a gap. If J_p were principal, write $J_p = xR_p$. Since $1 \in J_p$, there is $r \in R_p$ with $xr = 1$, so $rJ_p = R_p$ and $(r, rt^s) = R_p$. In the local ring, one of these two generators must be a unit. Positive valuation excludes rt^s, so r is a unit and $t^s \in R_p$, a contradiction. The ideal is therefore nonprincipal and rigid. The multiplicities $24p$ are distinct, so the family contains infinitely many distinct rings. □

9.2 Exact support and boundaries

The proof of Theorem 9.2 is independent of a finite sweep. As adversarial support, the implementation evaluated every $p = 2, 3, \ldots, 300$ [**MV**]. Exactly the predicted boundary values $p = 2, 3$ fail, while all 297 values $p \geq 4$ pass the seven identities, closure, generation, symmetry, invariants, and full-window/tail rigidity. The instances $p = 4, 5$ reproduce independently generated Boolean models byte-for-byte. Two corrupted positive controls are rejected. A separate auditor reconstructs all 297 positive membership hashes from the formula and performs fresh semantic checks at $p = 4, 5, 17, 73, 151, 300$. The campaign aggregate is

`81d5a8eb6cf2e848807323e3b0bdba58c464779d25cdc788cef027585540dce2`.

The independent-audit aggregate is

`eb2aaf17650ed99f4e220a43c53bdd8835c82688a37567bb154c30a1ae520ce9`.

The family remains outside the generalized-arithmetic-sequence positive class [**D**]. Membership of $4s + 1$ forces unit step, while membership of $4s + 3p$ would then force the actual gap $4s + p + 1$. Theorem 9.2 does not say that every model in the Boolean scaffold belongs to this family, or that $p = 4$ is a global threshold outside the displayed formula.

10 Reproducibility and trust boundary

All compact code, hypotheses, manifests, audits, and verdicts are in the public repository under

`problems/commutative-algebra/huneke-wiegand/`.

Heavy CNFs and proofs are retained outside Git with hashes committed in the manifests. The minimum-layer archive adds 12 fully accounted files and a reconstruction auditor. The proof toolchain is pinned as follows:

Component	Identity
CaDiCaL	reported 1.7.3; Ubuntu package 1.7.4-1
CaDiCaL binary SHA-256	`7b73df0a6d9cf3c751a1948300e5baff8e82c4d39bcd88f0c063b5f5cfb8b33e`
DRAT-trim commit	`2e3b2dc0ecf938addbd779d42877b6ed69d9a985`
DRAT-trim binary SHA-256	`92f0aa9575ed519d66a99b8b1b3dde6ece4618ae4c202a3a4b200265dda0aa7a`

The trusted base comprises the cited tree theorem, the finite reduction proved here, the small standard-library model checker, and DRAT-trim's verification of each proof against its exact CNF. The SAT solver need not be trusted for UNSAT. For SAT, the independently extracted model is checked without consulting solver internals.

11 Scope, attribution, and open questions

Son Pham retains discovery priority for the counterexample. Professor Craig Huneke's verification is independent external evidence. The contributions proved here are Frobenius minimality, uniqueness of the normalized pair at that minimum, and the separate infinite family in Theorem 9.2, together with their reproducible proof and certificate records.

Theorems 6.1, 7.2, and 9.2 do not classify all higher Frobenius values, prove a minimum multiplicity or embedding dimension across all counterexamples, or establish corresponding statements for arbitrary Gorenstein domains and modules. The following questions remain open:

1. How does the number of normalized rigid pairs grow above $F = 181$?
2. Is there a global minimum under multiplicity, embedding dimension, or number of minimal generators?

3. Which other Boolean-scaffold models lie on affine or residue-class families, and is the family in Theorem 9.2 maximal inside a natural Kunz face?
4. Which additional hypotheses on the endomorphism ring recover a positive theorem while excluding the public example?
5. Can the finite certificates be imported into a smaller formally verified checker?

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